

Module 03: Perception in Robotics

Learning Objectives

1. Define **Perception** within the context of Robotics.
2. Analyze how Perception interacts with other components of the Active Inference framework.
3. Apply specific constraints of Robotics to the formal definition of Perception.

Introduction

This module explores **Perception**. In the **Robotics** curriculum, we approach this topic with a focus on specific applications and theoretical depth appropriate for the audience. Perception is a critical component of the 8-part Active Inference spine, bridging the gap between Agents and Cognition.

Key Concepts

1. Perception as a Markov Blanket Boundary

How does Perception define the boundary between the agent and the environment?

2. Generative Models of Perception

What parameters involved in Perception must be optimized to minimize variational free energy?

3. Active Inference Dynamics

How does the process of Perception drive the perception-action loop?

Applications

In Robotics, we see Perception manifest in: * **Specific Example 1:** A bat-inspired robot uses biomimetic sonar with dynamically deformable pinnae (ear-shaped acoustic reflectors) to implement active perception -- by reshaping its ear geometry during echolocation, the robot modulates the directional sensitivity pattern of its ultrasonic receivers, actively sampling acoustic space to reduce uncertainty about obstacle geometry in dark environments, directly paralleling how biological bats adjust pinna shape to extract 3D spatial information from echo returns. * **Specific Example 2:** A lateral-line-inspired underwater robot equipped with arrays of pressure-sensitive artificial neuromasts along its body perceives hydrodynamic flow fields the way fish do; the distributed sensor array provides a likelihood function over nearby object positions and water currents, and the robot's generative model inverts this flow-field data to infer the presence, distance, and velocity of neighboring objects even in zero-visibility turbid water where vision-based perception completely fails.

Conclusion

Understanding Perception allows us to better model complex adaptive systems. In the next module, we will expand on this foundation.